

Ethanol Blending in Gasoline – Hungary

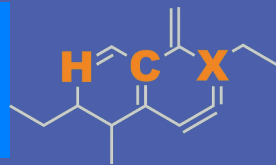
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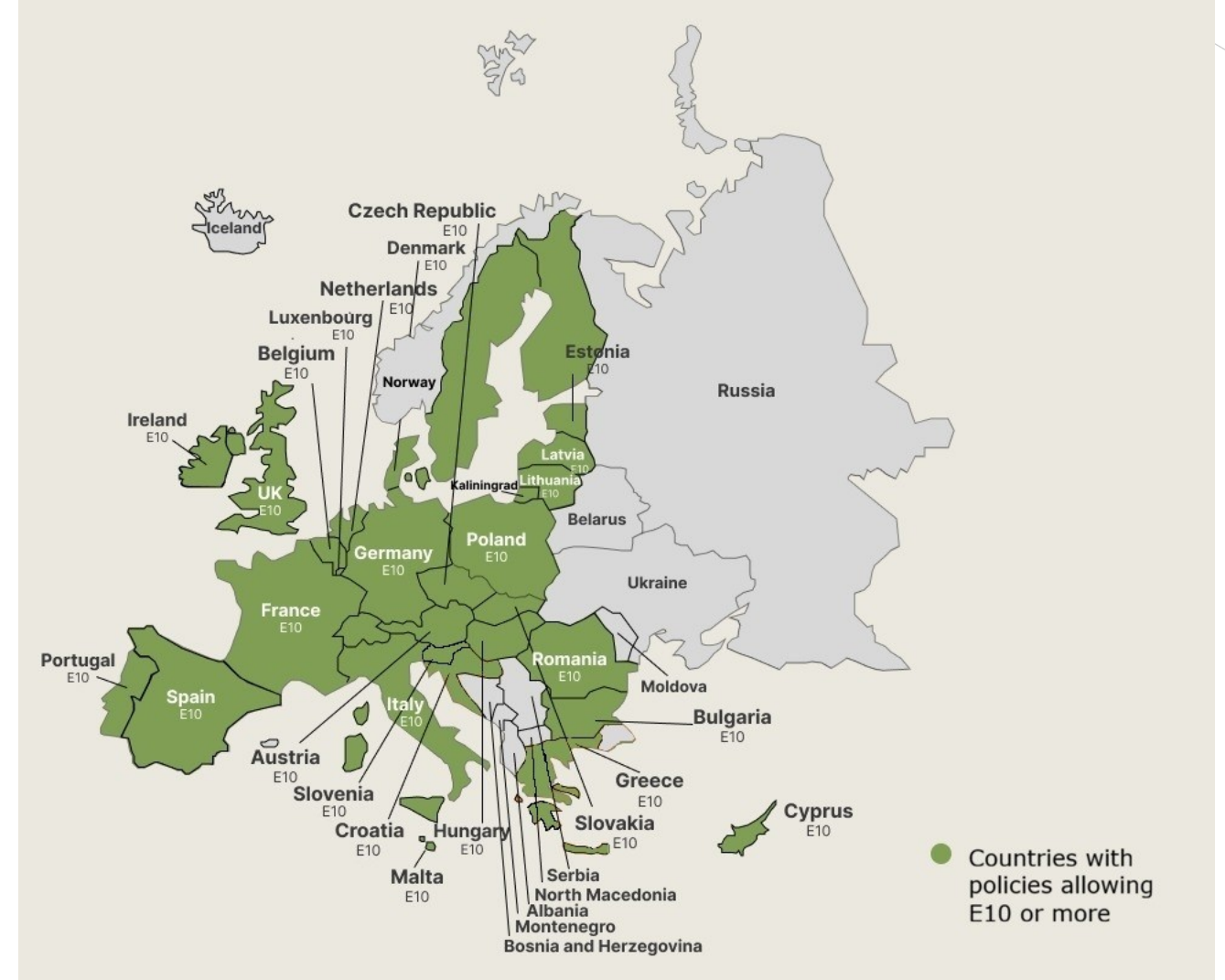


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Ethanol Blending in Europe

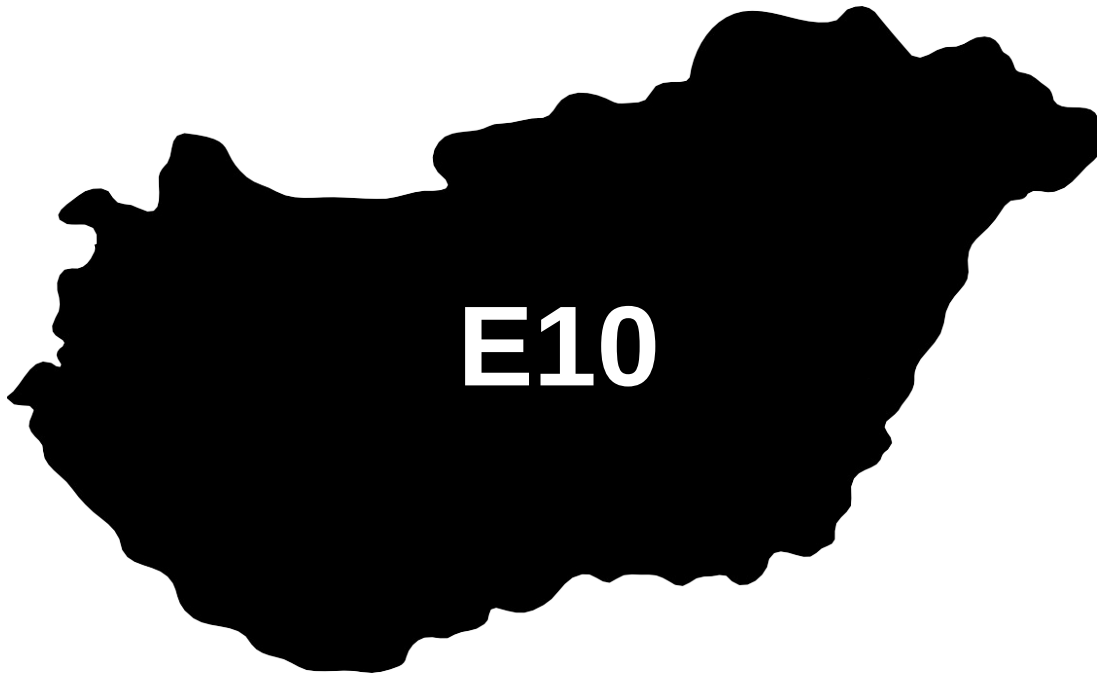
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

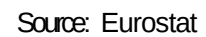
Ethanol Gasoline Blending in Hungary



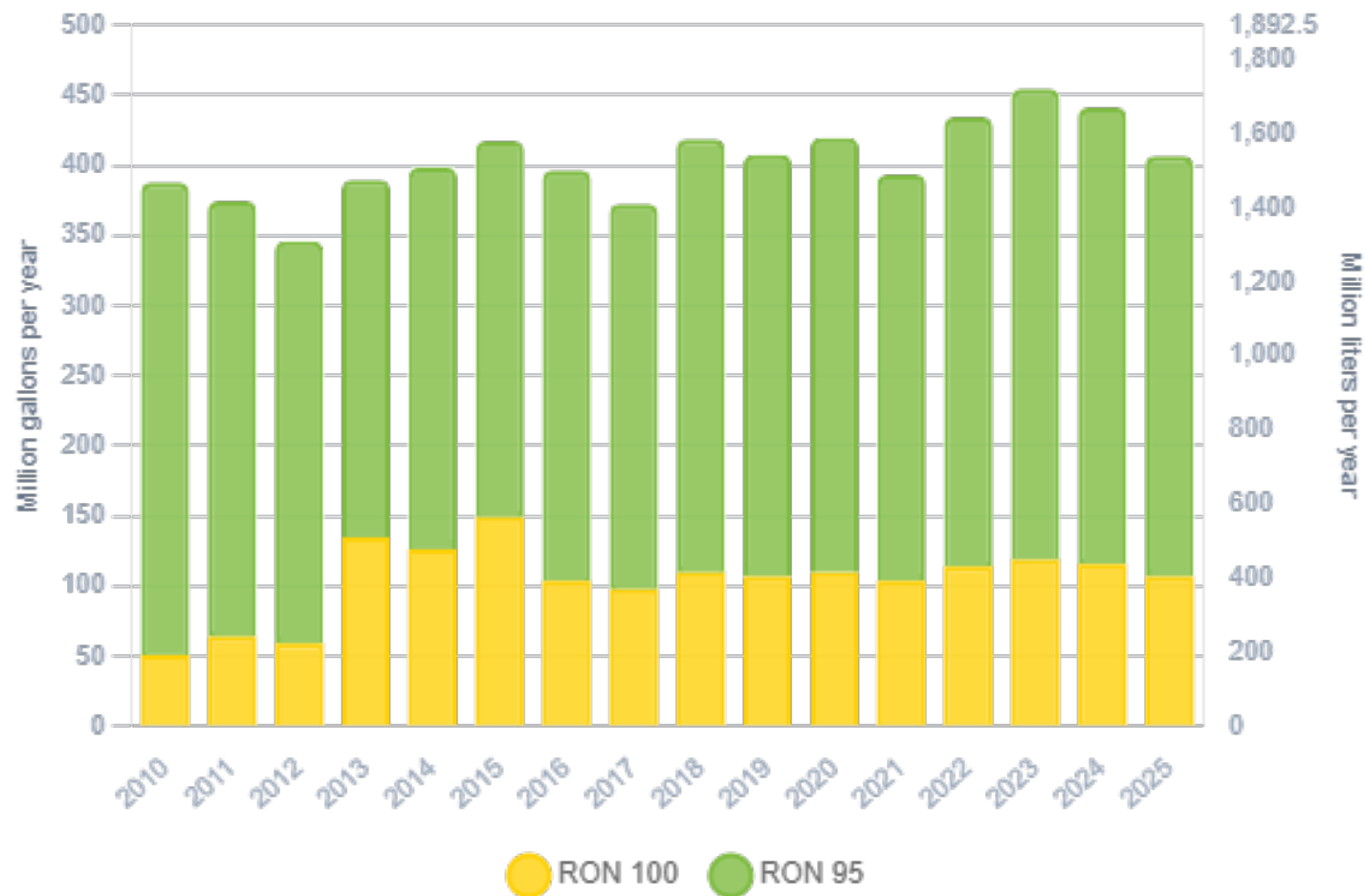
In 2025, gasoline consumption in Hungary was 1,908 million liters (504 million gallons) and growth rate was 2.2%. Gasoline production and net imports were 1,438 and 473 million liters (380 and 125 million gallons) correspondingly. Hungary complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.4 RON.

In 2025, ethanol consumption in Hungary was 113 million liters (30 million gallons) representing 6% of gasoline demand. Ethanol production and Net Imports were 531 and -418 million liters (140 and -110 million gallons) correspondingly. Ethanol source is mainly Corn based.

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a double bond between carbons 1 and 2, with substituents H, C, and X at the 1-position, and a methyl group at the 4-position.

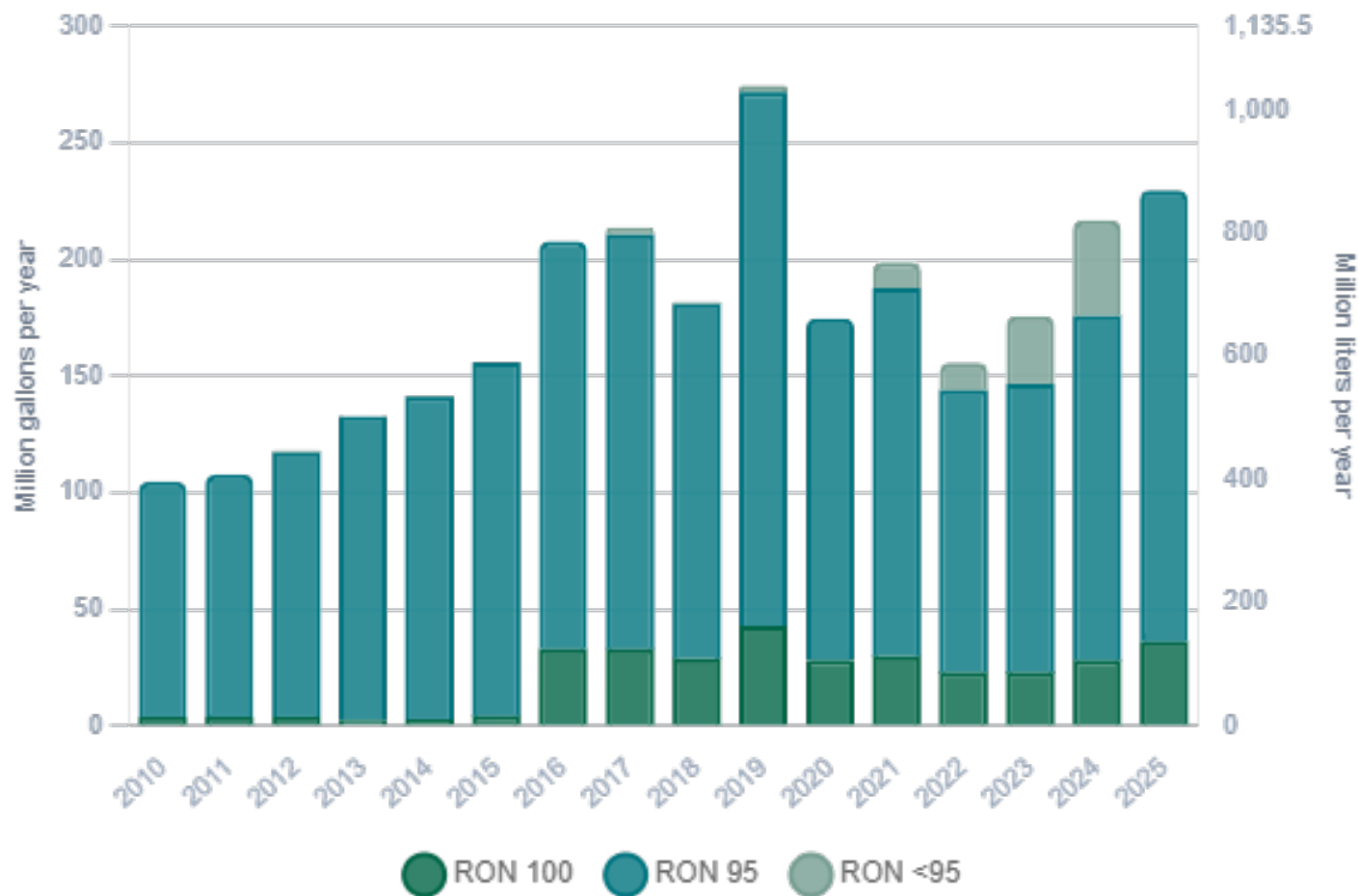


Gasoline Production in Hungary



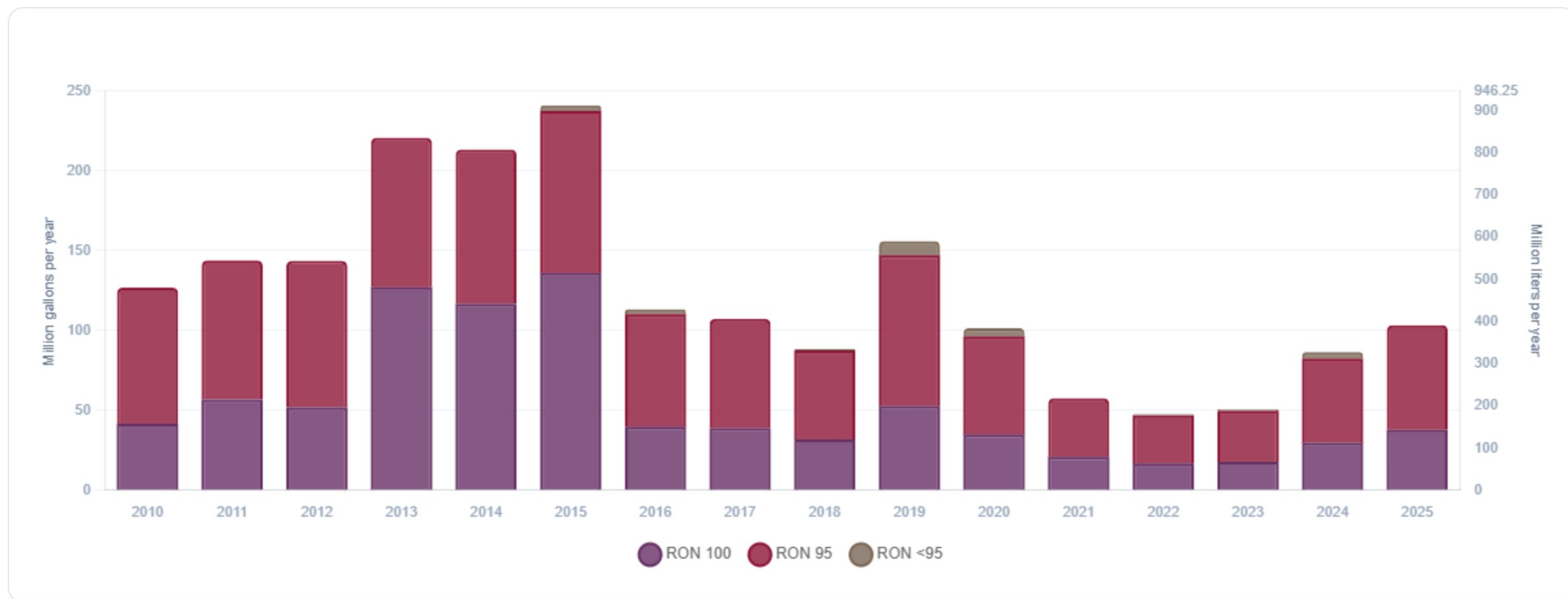
Source: Eurostat

Gasoline Imports in Hungary



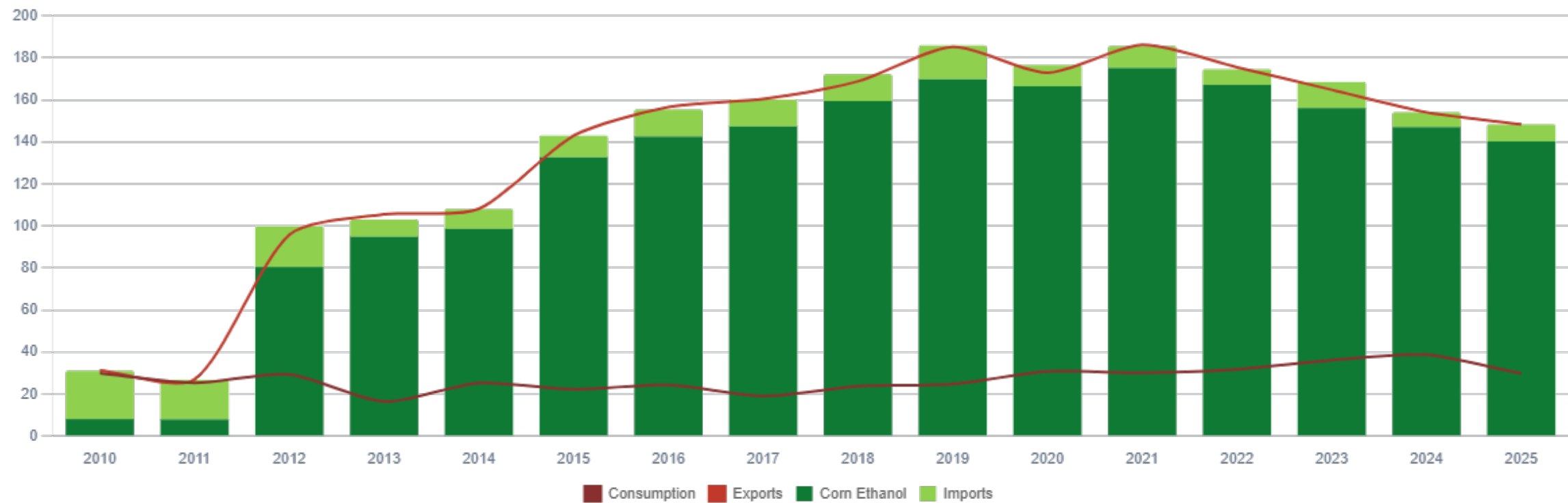
Source: Eurostat

Gasoline Exports in Hungary



Source: Eurostat

Ethanol Balance in Hungary



Source: Eurostat

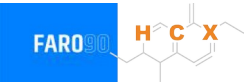
Gasoline Quality in Hungary

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	6.1							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

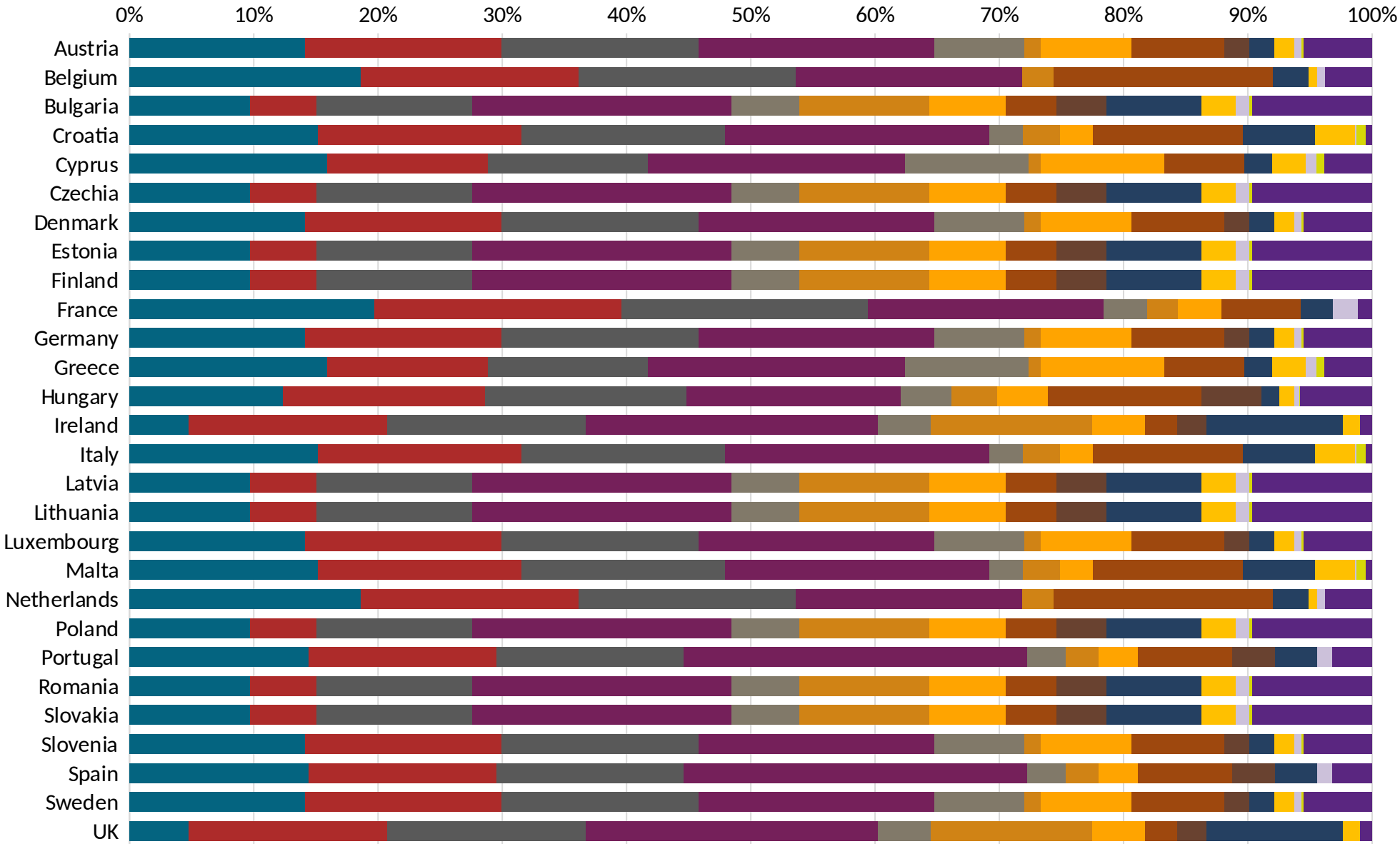
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



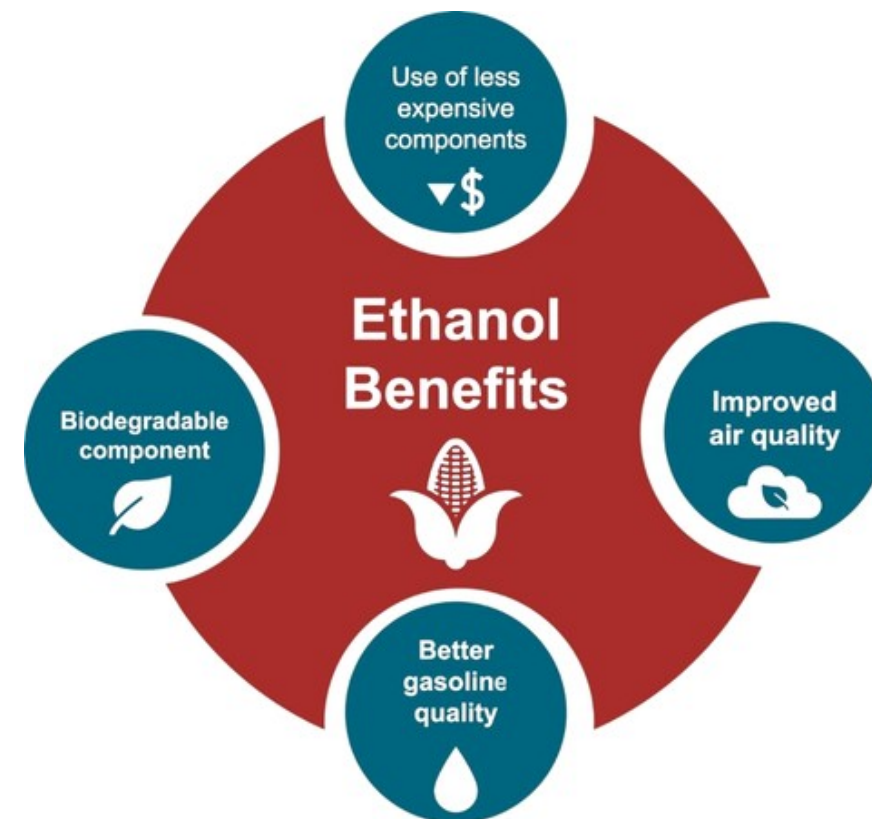
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

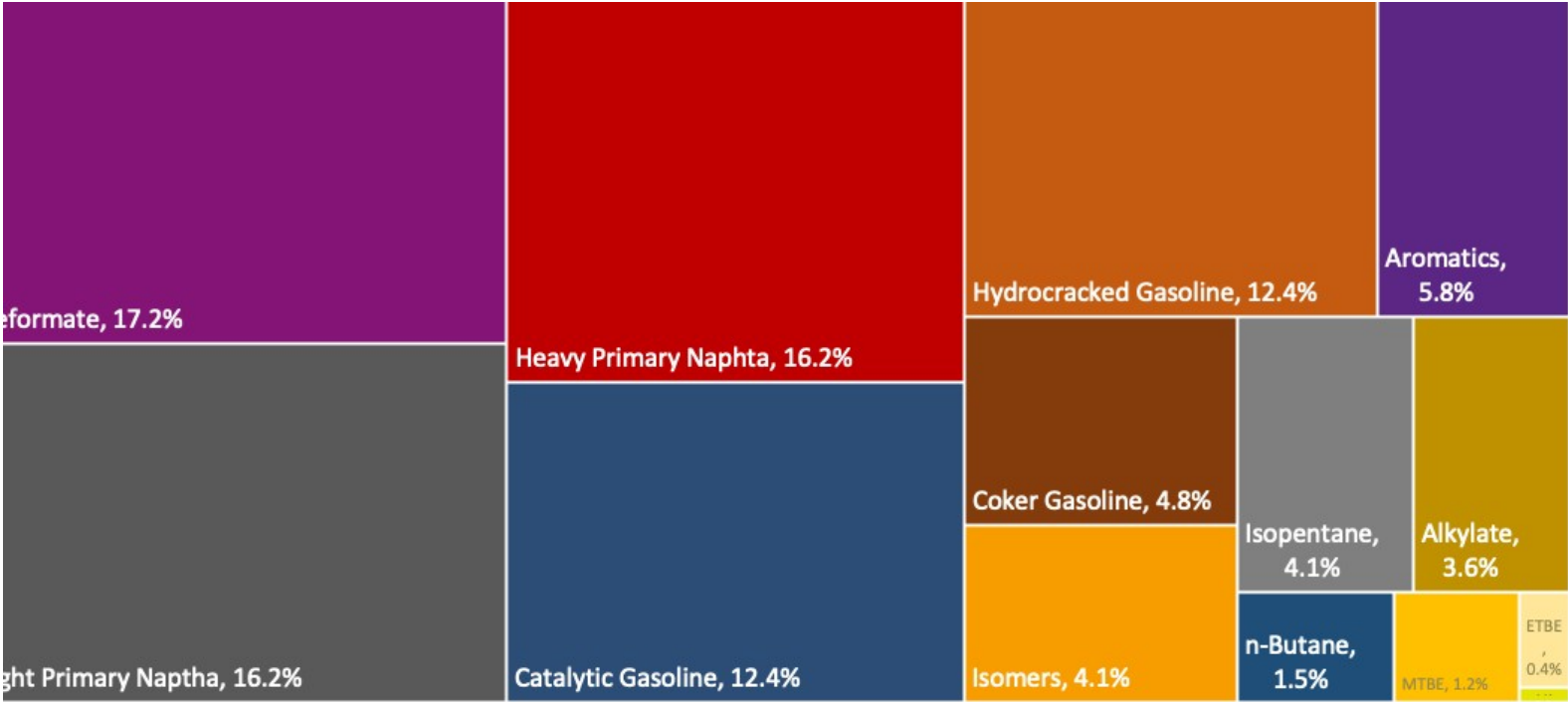
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

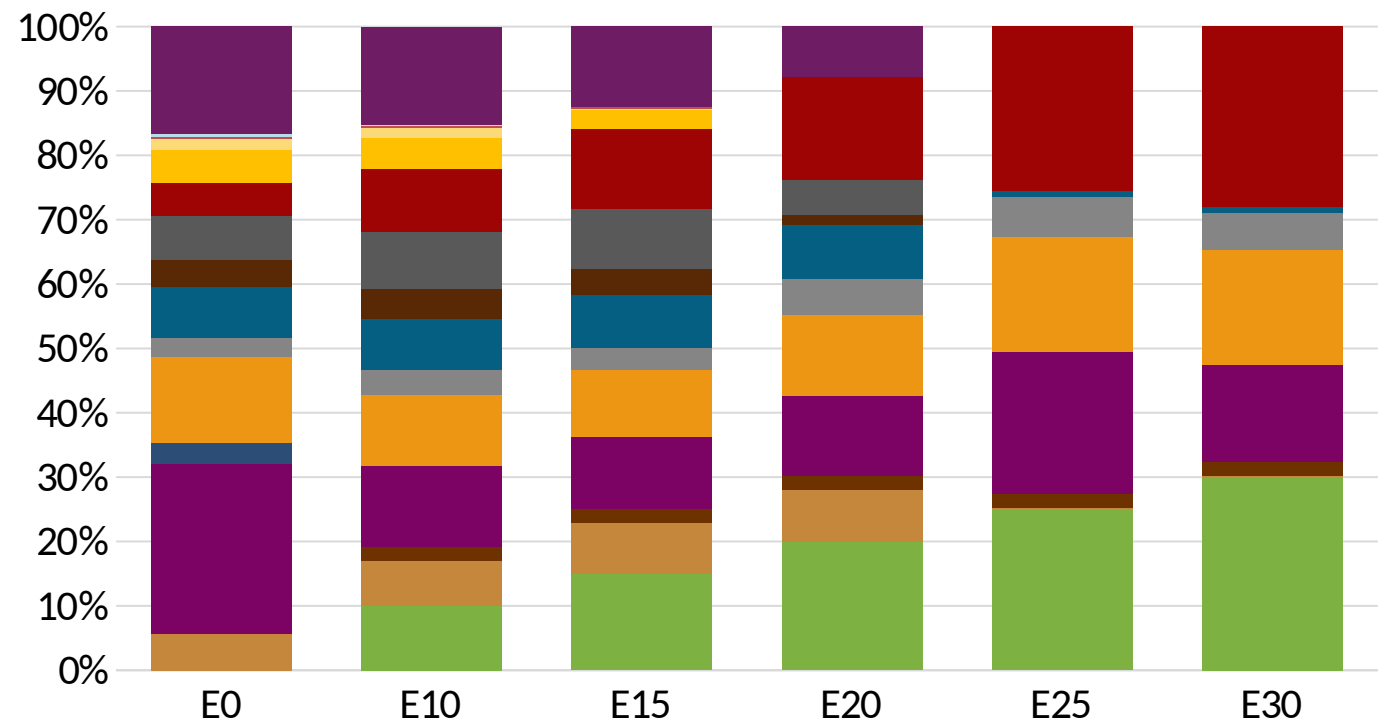


Hungary Available Blending Components



TAME,
0.1%

Hungary – Gasoline 95 – Constant Octane

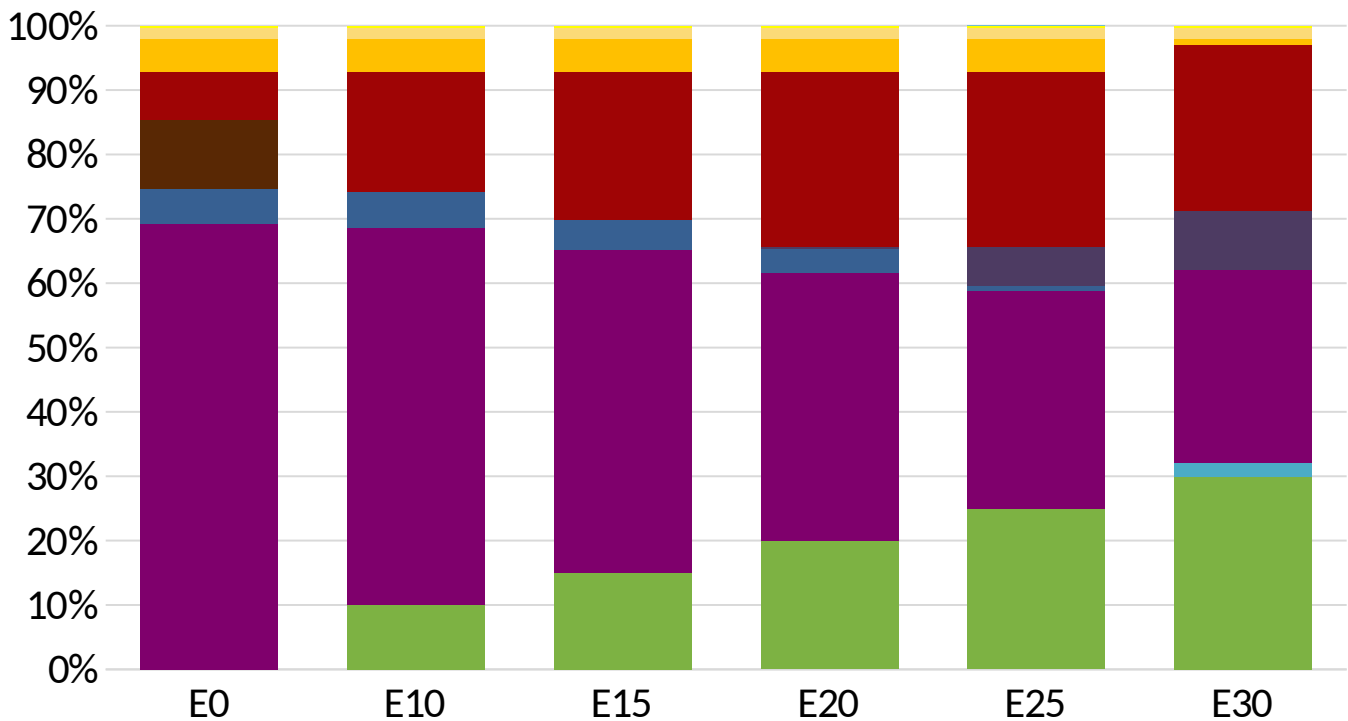


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.094	\$ 1.965	\$ 1.896	\$ 1.818	\$ 1.675	\$ 1.638

Source:
Faro90

Hungary - Gasoline 98 - Constant Octane

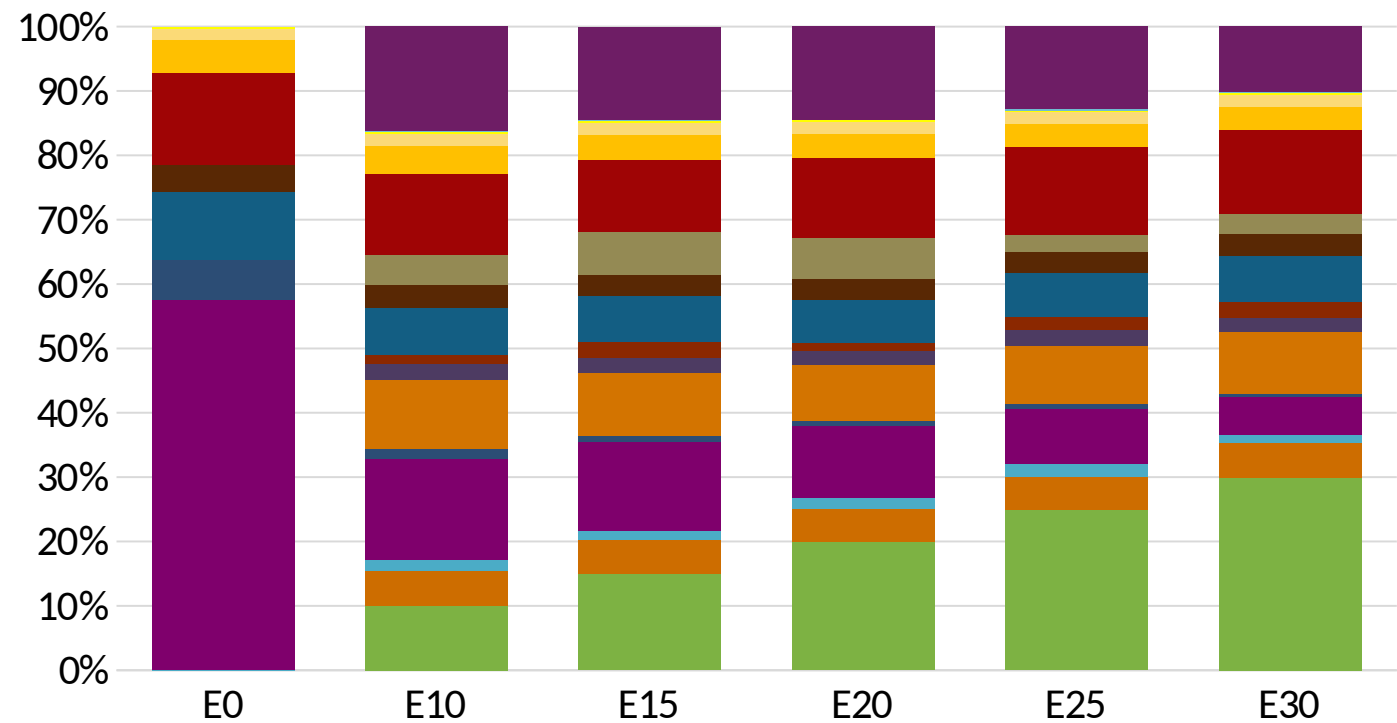


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.126	\$ 2.005	\$ 1.936	\$ 1.861	\$ 1.784	\$ 1.720

Source:
Faro90

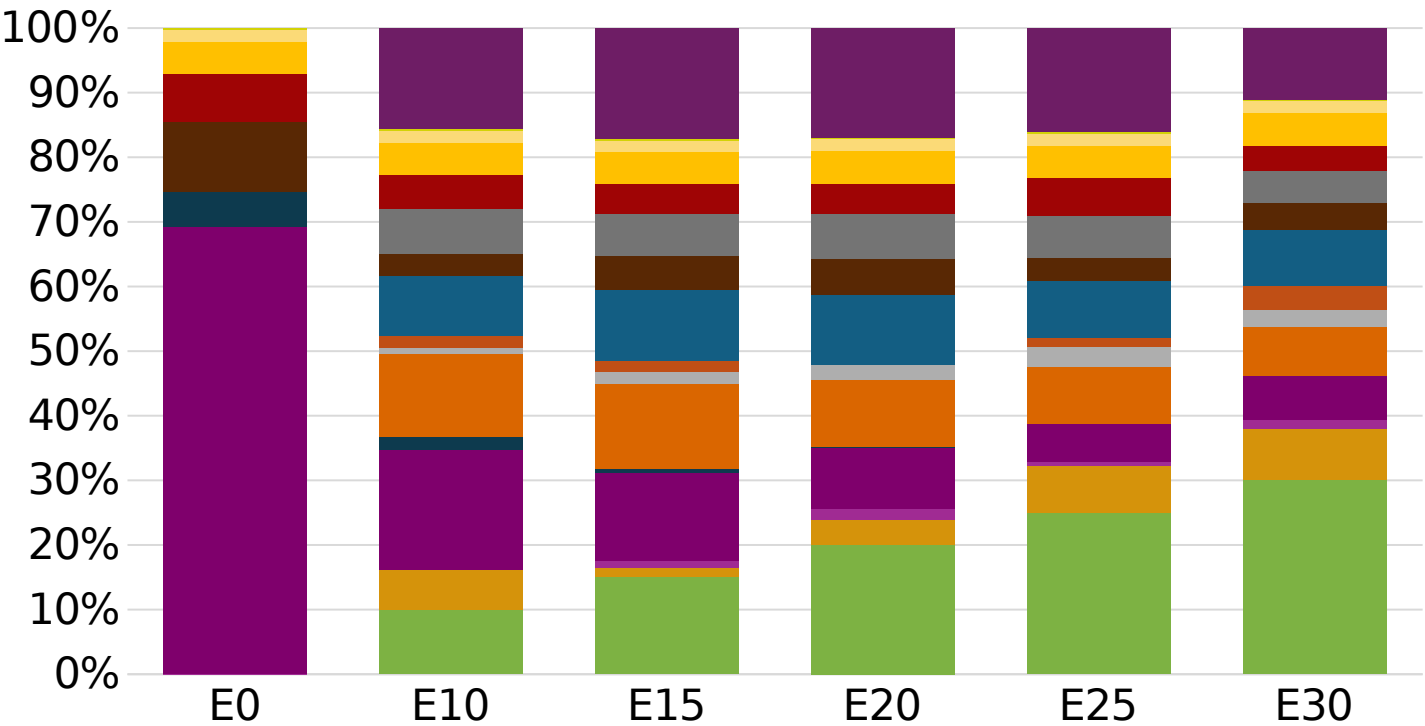
Hungary – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.7	97.2	98.8	100.3	101.8
Price (US\$/gal)	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995

Hungary – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

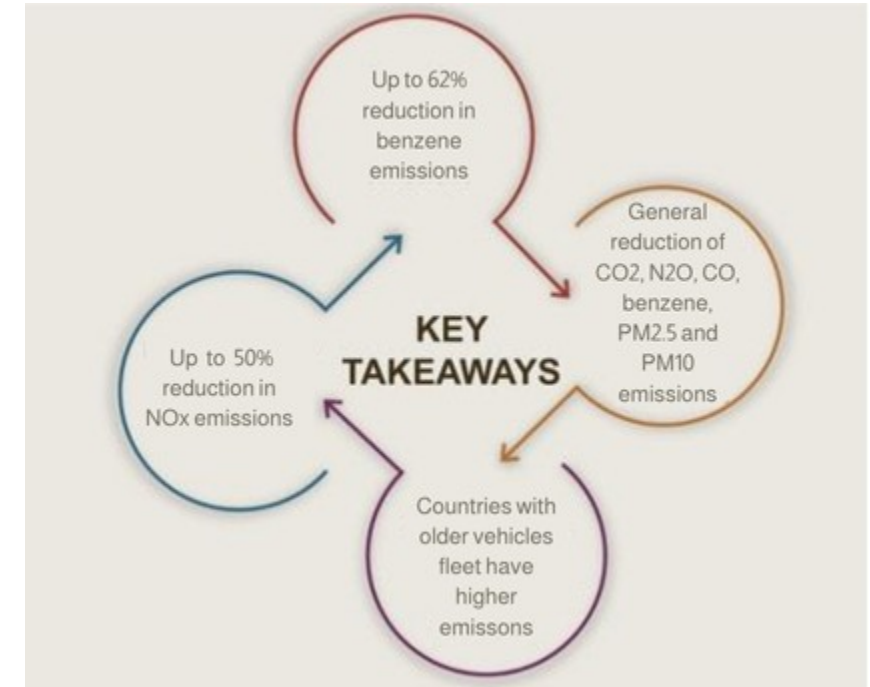
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

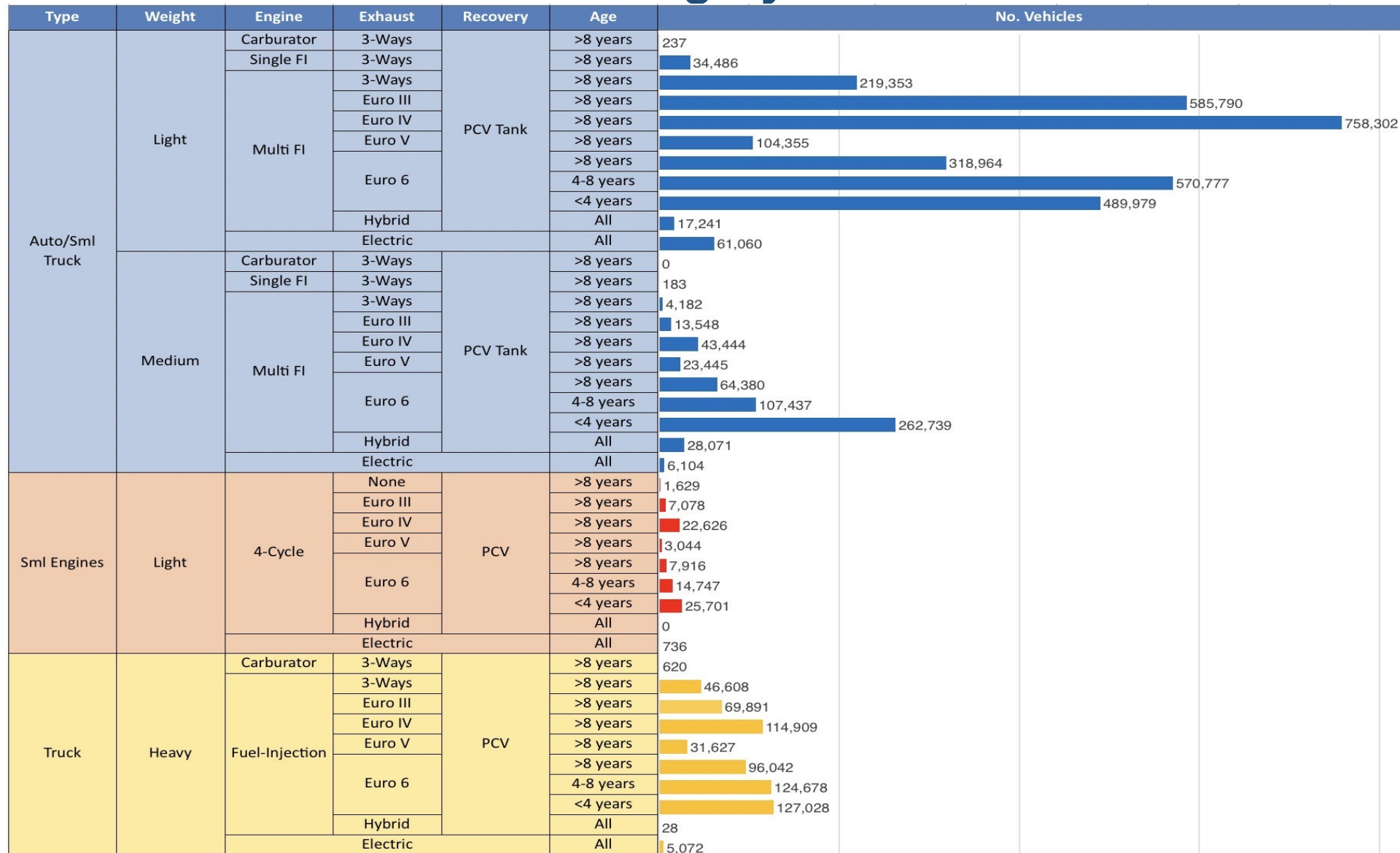
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Hungary

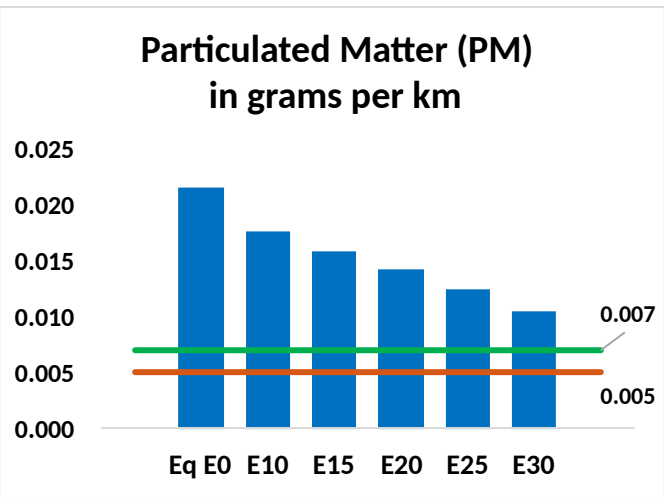
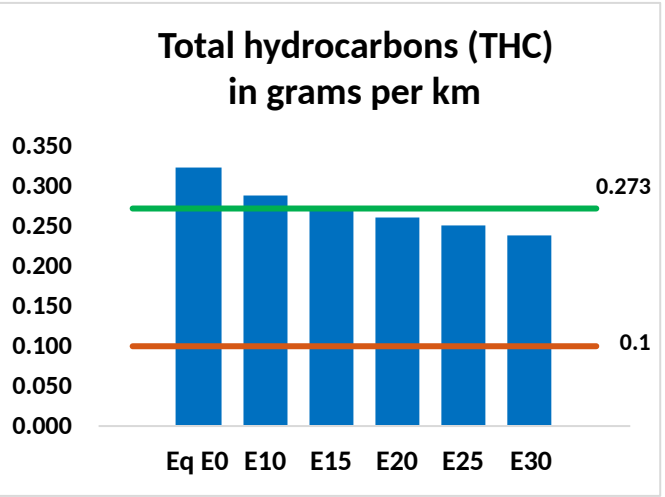
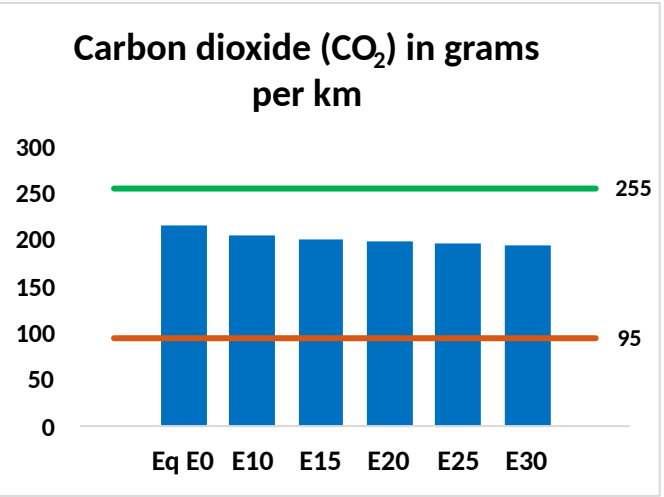
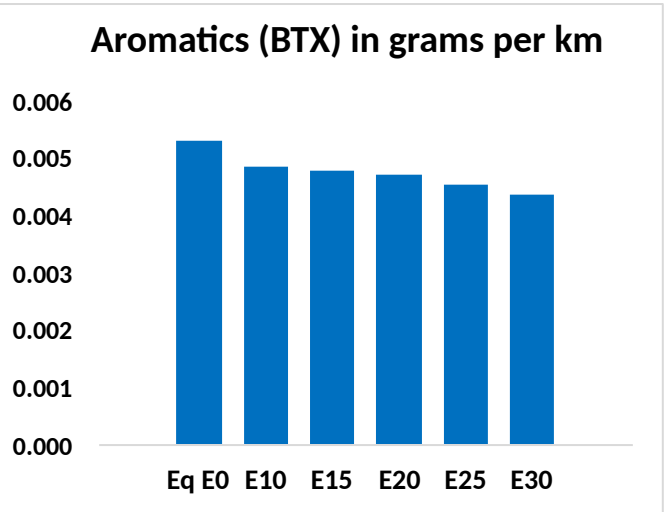
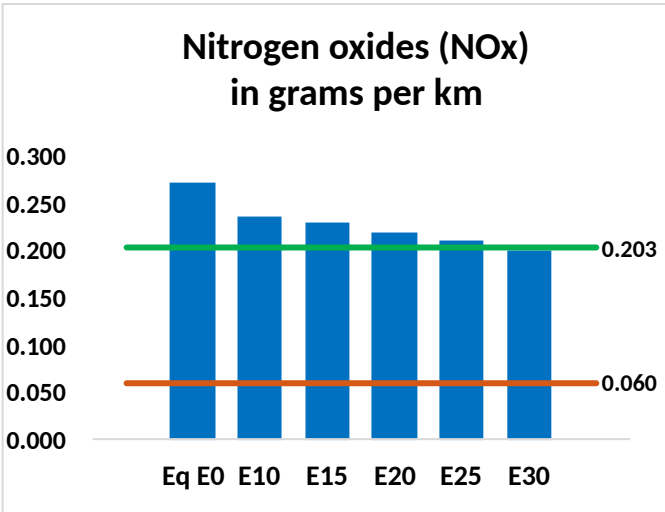
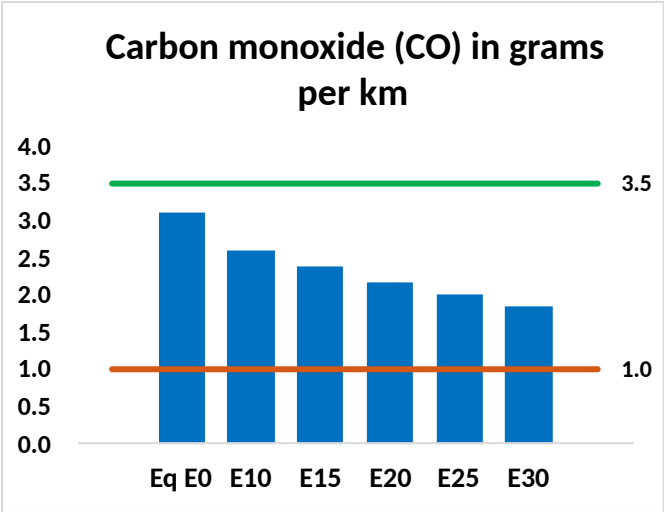
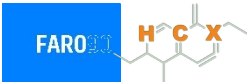


Vehicle Fleet: 4,414,058
Gasoline Vehicle Fleet: 2,379,575

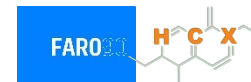
Source: Eurostat, ACEA

Hungary – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Hungary – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	179,554	164,907	150,260	138,093	125,773	116,277	105,990	-16%	-30%
CO2	12,474,387	12,153,617	11,850,668	11,612,407	11,494,506	11,431,684	11,220,966	-5%	-8%
VOC	18,773	17,724	16,676	15,894	15,142	14,575	13,780	-11%	-19%
NOx	15,810	14,598	13,783	13,376	12,711	12,195	11,584	-13%	-20%
SOx	141	131	120	111	102	93	83	-15%	-28%
NH3	4,231	4,189	4,147	4,167	4,214	4,247	4,274	-2%	0%
N2O	193	192	191	192	196	198	200	-1%	2%
CH4	4,091	4,091	4,091	4,152	4,242	4,320	4,394	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	163	157	152	149	146	144	141	-7%	-10%
Acetaldehyde	351	403	592	901	1,227	1,403	1,657	68%	249%
Formaldehyde	1,103	1,073	1,250	1,468	1,538	1,702	1,852	13%	39%
Benzene	310	297	282	277	275	263	255	-9%	-11%
PM2.5	730	664	598	539	482	421	357	-18%	-34%
PM10	523	476	428	386	345	301	255	-18%	-34%

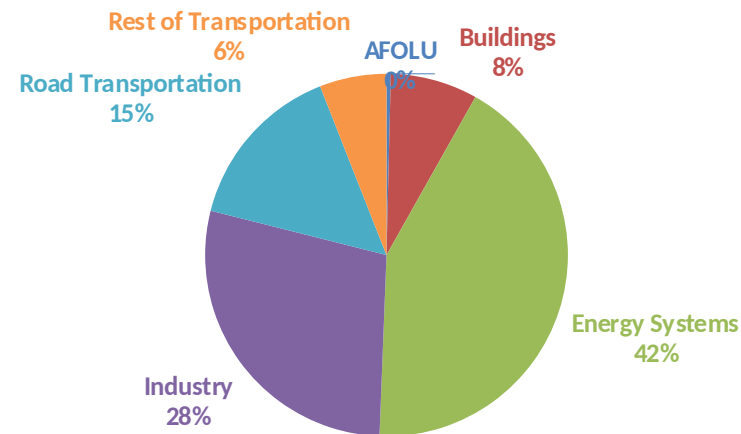
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

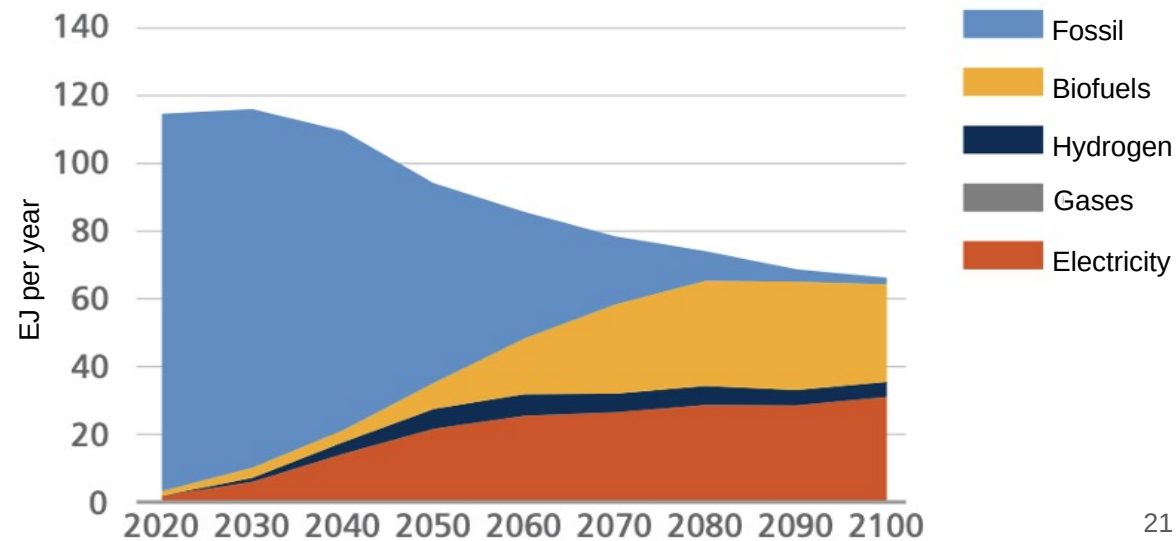
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

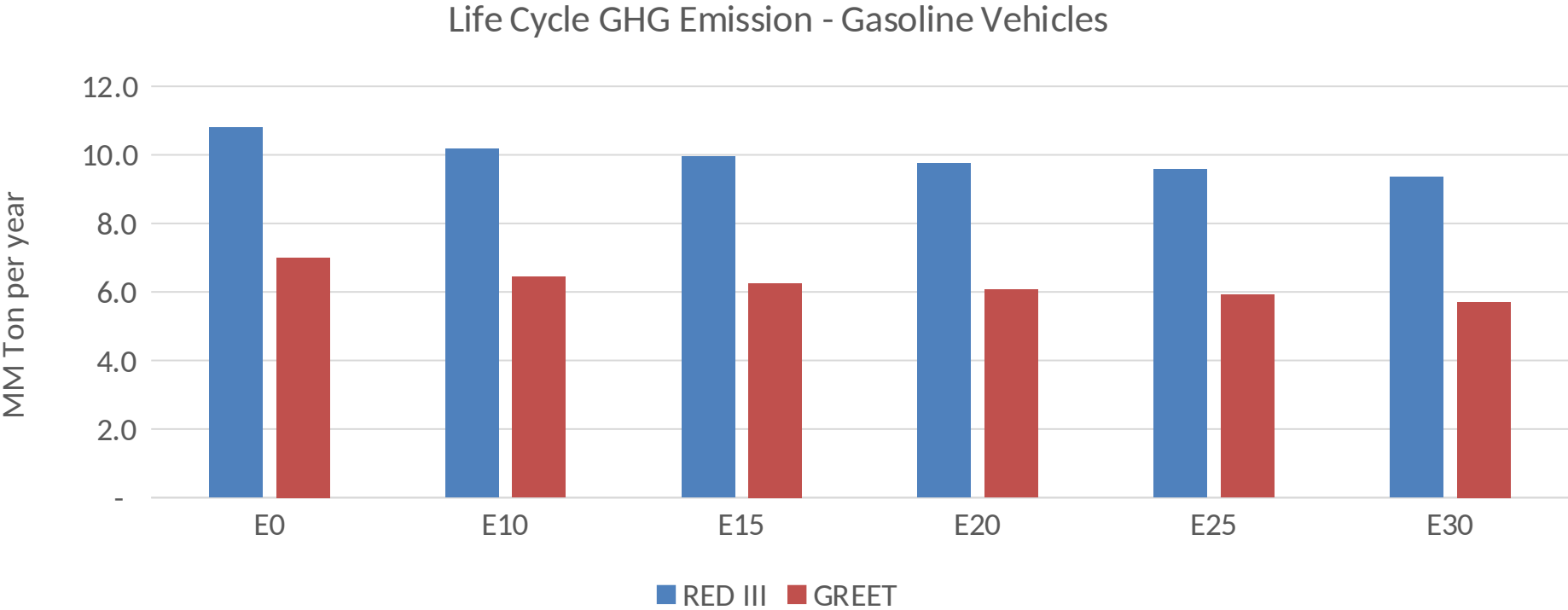
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Hungary – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	64.6
Target @ 2035 (MMT/year)	36.4
Delta	28.2